

HUGO HU

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EDUCATION

Stony Brook University

Stony Brook, NY

Expected May 2030

- B.S. Computer Science, Sophomore Standing
- Honors College, Honors in Computer Science Program

Stuyvesant High School

New York, NY

- Excellence in Mathematics Award, AIME Qualifier (3x), CEMC Euclid/Fermat School Winner June 2026
- Relevant Coursework: Linear Algebra, AP Calculus BC, Cybersecurity, AP Physics C GPA: 97.64/100

TECHNICAL SKILLS

Languages: Python, Java, C#, C, SQL, MATLAB

Tools: Git, MongoDB, Supabase / PostgreSQL, KiCAD, LaTeX, HTML/CSS

Frameworks: Next.js, React.js, Express.js, Node.js

AI/Development: Agentic AI, LLM APIs, Tool Calling, Structured Outputs

PROFESSIONAL EXPERIENCE

Stony Brook Campus Residences

Stony Brook, NY

IT and Programming Intern

September 2026 - Present

- Migrate legacy asset-tracking applications into unified React/Express.js web interface, consolidating re-supply and service workflows; develop a Google Sheets synchronization pipeline

Hack Club

Shelburne, VT and Remote

Hardware Engineering Intern

May 2022 - March 2024

- Led design and fabrication of production PCBs for Blot (2D CNC drawing machine) and Sprig (handheld gaming console), overseeing process from prototype to fab; Sprig reached a 600+ unit production run
- Published open-source reference designs adopted by the maker community (NTAG NFC Tag, USB-C Hub)
- Mentored dozens of high schoolers with design review and expertise through first-time PCB design

Logistics and Operations Intern

July 2021 - August 2023

- Built a mailing address database using USPS Intelligent Mail Barcode, enabling no-cost trackable postage
- Reduced domestic shipping costs by up to 60% and international costs by up to 80%

RESEARCH EXPERIENCE

Memorial Sloan Kettering Cancer Center

New York, NY

Medical Physics Student Intern

May 2025 - November 2025

- Fine-tuned and validated a CNN-based deformable image registration (DIR) model for aligning cone-beam CT scans to planning CTs for lung cancer treatment monitoring, achieving validation DICE score over 0.75
- Prototyped a multi-head attention DIR architecture, reaching parity with CNN and VoxelMorph baselines
- Ran training and inference on HPC clusters and configured CUDA environments for training compatibility
- Wrote MATLAB and Python code to preprocess and validate training datasets

Medical Physics Summer Intern

July 2024 - August 2024

- Wrote C#, Python preprocessing code for head/neck cancer segmentation training for treatment tracking
- Leveraged HPC clusters and IBM LSF (bsub) for job scheduling

PROJECTS

Component Inventory Manager | Next.js, Claude API, Supabase | GitHub

2026

- Implemented a full-stack Next.js/Supabase system for electronic components with inline editing, flexible part search, and thermal-printable QR-coded asset labels
- Built a Claude invoice import pipeline using structured outputs with human review before database writes
- Designed a tool-using chat agent for natural-language search and batch labeling, with cost-aware routing

Personal Photography Portfolio | Next.js, NextAuth, Cloudinary CDN | GitHub

2026

- Designed a performant Next.js photography site with Cloudinary CDN delivery, EXIF extraction, and category-based filtering, cutting image load times and simplifying browsing across collections
- Built a secure authenticated admin dashboard (NextAuth, Bcrypt) backed by Supabase PostgreSQL, enabling non-technical content updates without redeploying the site